

FINITE-SAMPLE CERTIFICATES FOR ADAPTIVE STOPPING IN COUNT-EXCHANGEABLE BINARY PASS-COUNT REPLAY

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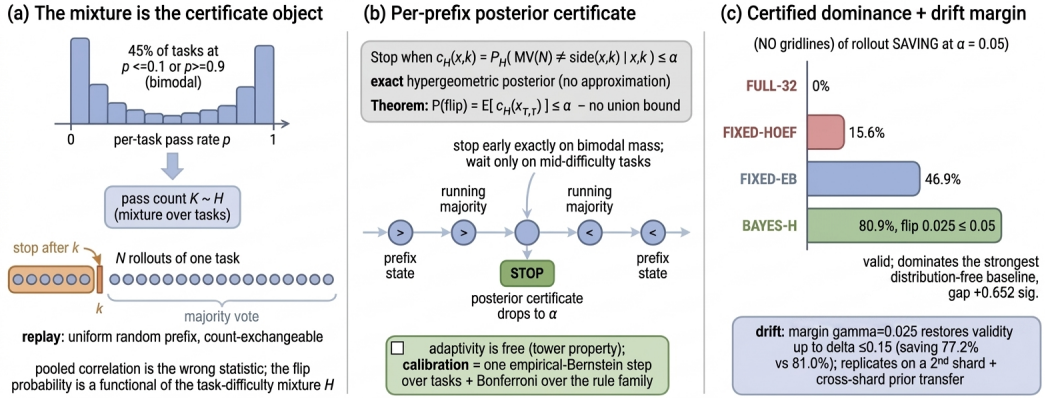
Paper under double-blind review

ABSTRACT

We study an analysis model for repeated model rollouts: a *count-exchangeable binary pass-count replay* in which a task’s N outcomes are available only through its pass count. In that model, we give a *mixture-posterior stopper* (BAYES-H) whose per-problem conditional flip certificate is exact: given a FIT-frozen prior H over per-task pass counts, stop as soon as the posterior probability that the full- N binary pass-count decision will disagree with the current prefix decision drops below α . By the tower property the replay-population flip rate of any such prefix-measurable rule is bounded by α *without* a union bound over stopping times (adaptive is free). Under independent i.i.d. FIT/CAL task sampling, the population layer adds an empirical-Bernstein CAL step with Bonferroni control of the pre-declared frozen rule family. On 11,607 math problems with $N = 32$ rollouts each (OpenMathReasoning), selected and evaluated under a strict FIT/CAL/TEST problem split with a single held-out readout, BAYES-H has an 81% replay rollout-count reduction at realized flip rate $0.025 \leq \alpha = 0.05$ and 73% at $\alpha = 0.02$, dominating the strongest distribution-free baseline (Hoeffding-fixed- k : 16% at $\alpha = 0.05$) with a statistically significant paired gap, and the strongest mixture-aware fixed budget (empirical-Bernstein fixed- k : 47%). An ESC-style window heuristic with an 86% replay count reduction exceeds 0.05 ($0.058 > 0.05$). Under pre-fixed synthetic ordered-drift mechanisms (front-loading, linear trend, adversarial block swap), BAYES-H is the most robust stopper in the reported harness. The displayed certificate-margin thresholds are exploratory descriptive sensitivity results, not frozen-selection or confirmatory claims. The supplied replay readout replicates on a disjoint second shard; transferring one fitted calibration mixture across shards (TV 0.042) moves its reported quantities by ≤ 0.4 points. This is deterministic fitted-mixture sensitivity, not a population-transfer certificate. Two further model-carrier pairs bracket the mechanism: on OpenR1-Math-220k (DeepSeek-R1, $M = 2$ per problem) the certificate attains the replay risk-count frontier exactly when the mixture structure allows, and never stops under the stated replay certificate when it cannot; on RLVE (Qwen3-4B, $N = 8$ per problem, verifiable environments) it lands at the intermediate point the mechanism predicts — 51–58% replay count reduction at $\alpha \in \{0.10, 0.05\}$, 45–48% at $\alpha \in \{0.02, 0.01\}$, certified at *every* level down to $\alpha = 0.01$ where *no* distribution-free replay budget is certifiable. Every reported flip is disagreement between corresponding binary pass-count replay decisions, and every saving is $1 - \bar{k}/N$; these artifacts do not measure a delivered-answer majority, gold correctness, generated tokens, latency, cancellation, or ordered online deployment.

1 INTRODUCTION

Self-consistency (Wang et al., 2023) replaces one sampled answer by a majority vote over N samples. Adaptive variants reduce or reallocate sampling: ASC (Aggarwal et al., 2023) adapts a per-problem count with a lightweight stopping criterion, ESC (Li et al., 2024) proposes early stopping for self-consistency, and MARS (Chen et al., 2026) stops parallel traces using a conservative bound on future vote movement. For a count-only replay analysis, two deployment factors remain outside



Treat across-task heterogeneity as the certificate object (a): stop when the exact posterior flip probability drops to α (b) -- adaptivity is free and the result saves 81% of rollouts where the strongest distribution-free baseline saves 16% (c).

Figure 1: **Conceptual overview with artifact-checked quantitative labels.** (a) Treat across-task heterogeneity as the certificate object: per-task pass rates are strongly bimodal, so the flip probability is a functional of the difficulty mixture H , not of a pooled correlation. (b) BAYES-H stops when the exact posterior flip probability $c_H(x, k)$ drops to α ; by the tower property adaptivity is free (no union bound), and calibration is one empirical-Bernstein step over tasks. (c) At $\alpha = 0.05$ BAYES-H has an 80.9% replay rollout-count reduction (realized flip 0.025), dominating the strongest distribution-free baseline (FIXED-HOEF, 15.6%) and the strongest mixture-aware fixed budget (FIXED-EB, 46.9%); a certificate margin controls flip only in stated synthetic drift mechanisms, and all results replicate on a disjoint second shard. Panel (c)’s displayed numbers are rounded reproductions of the named replay artifacts, not measurements generated by the figure. The flip and count quantities are replay artifacts, not online correctness or operational-cost measurements (REPRO_README.md).

its evidence: (i) within-task rollouts may be correlated; and (ii) chronological generation order may drift, breaking an exchangeability-based certificate.

We make two structural observations that reframe the problem. First, on real carriers the dominant deviation from i.i.d. is not latent within-task correlation but *across-task heterogeneity*: per-task pass rates are strongly bimodal (Section 3), and the flip probability a stopper must control is a functional of the *mixture* of per-task difficulty, not of a pooled correlation coefficient. Pooled second-moment summaries can be simultaneously over-conservative and uncalibrated (Section 3). Second, once the target is the mixture, the correct object is a *per-prefix conditional certificate*: the posterior probability, given the observed prefix, that the full- N majority disagrees. This object is exactly computable (hypergeometric posterior), its fixed-budget expectation $\mathbb{E}_H[f_K(k)]$ is monotone in the prefix length (Lemma 1), and — because the tower property collapses the stopping-time dependence — it certifies any adaptive rule at level α with no union bound (Theorem 1).

Contributions.

1. **Framework.** A *binary pass-count replay* model, not an ordered online or answer-aggregation model, in which both within-task correlation and across-task heterogeneity are folded into a mixture H over per-task pass counts; the certificate target is the replay flip probability under H (Section 3).
2. **Method.** BAYES-H: stop when the posterior conditional flip probability $c_H(x, k) \leq \alpha$. Exact certificates (Lemma 1, Theorem 1); adaptivity is free — no union bound, no e-process tuning (Section 4).
3. **Legitimate selection.** Under the stated i.i.d.-task and FIT/CAL-independence assumptions, a FIT-frozen, pre-declared rule family (budget $\times$ level for fixed- k ; level for BAYES-H) is selected on CAL with simultaneous empirical-Bernstein bounds (Maurer & Pontil, 2009) plus Bonferroni. The fixed benchmark TEST split is read once and reported only descriptively (Section 5).
4. **Empirical replay result.** On 11,607 OpenMathReasoning problems, 80.9% replay rollout-count reduction at realized flip 0.025 ($\alpha = 0.05$), vs 15.6% for the strongest distribution-free

baseline at the same level; ESC-style window stopping is invalid ($0.058 > 0.05$). Gains are paired-CI significant (Section 6). Two further carriers with different models (DeepSeek-R1 at $M = 2$; Qwen3-4B at $N = 8$ on verifiable environments) show the certificate attains the replay risk-count frontier (stop when the replay certificate permits, otherwise not) and interpolates with the resolvable mixture structure, i.e. the method is carrier-aware rather than a generic stop-early trick (Section 6).

5. **Honest boundaries.** The public carrier gives per-task pass *counts*, not ordered rollouts; it supports a random-prefix (count-exchangeable) analysis only as a model assumption and cannot validate ordered online deployment. We therefore bound the failure domain under three pre-fixed synthetic ordered-drift mechanisms; displayed margin thresholds are exploratory/descriptive (Section 7).

2 RELATED WORK

Adaptive sampling for voting. ASC (Aggarwal et al., 2023) adapts a per-problem sample count using a lightweight stopping criterion, and ESC (Li et al., 2024) proposes early stopping for self-consistency. We do not establish a finite-sample disagreement-with-full-vote guarantee for either; on our supplied heterogeneous replay carrier, the instantiated ESC-style rule exceeds 0.05 (Table 1). MARS (Chen et al., 2026) is a parallel trace-stopping method that bounds future vote movement; our work is instead a sequential count-exchangeable replay certificate. Best-of- n and budget-forcing methods (Snell et al., 2025) tune compute without stopping certificates.

Conformal and anytime inference. Conformal prediction (Vovk et al., 2005) gives marginal coverage under exchangeability; confidence sequences (Howard et al., 2021) give anytime-valid bounds for i.i.d. means. Our certificate is neither: it is an *exact posterior flip probability* under a calibrated discrete prior, valid conditionally per prefix (hence adaptivity is free), with the mixture estimated from calibration problems under an empirical-Bernstein bound (Maurer & Pontil, 2009). The Bonferroni control of a small pre-declared rule family is deliberately conservative and transparent.

Dependent sampling. Design-effect corrections (Kish, 1965) shrink effective sample size by an intraclass correlation; we show pooled correlation is the wrong functional here (Section 3). Beta-binomial and correlated-Bernoulli models (Skellam, 1948) capture overdispersion but target estimation, not stopping certificates for the flip event.

Feature boundary. Table 2 makes the source-verified distinction compact: the proposed object is a specified count-exchangeable full-vote *replay* endpoint with a calibrated-mixture posterior flip certificate. It is not a general ordered-online early-stopping, conformal, or anytime-inference guarantee.

3 MODEL: COUNT-EXCHANGEABLE REPLAY

A task’s N rollout outcomes are summarized by the pass count $K \in \{0, \dots, N\}$. Across tasks, $K \sim H$, a mixture capturing both task heterogeneity and any within-task correlation: the probability a fresh rollout passes, marginally, is $\mathbb{E}_H[K]/N$, and the joint law of the N outcomes is exchangeable *within* a task given its realized count. Analysis model (*replay*): the stopper observes a uniform random prefix drawn without replacement from the task’s N outcomes — i.e., the first k draws are exchangeable with the remaining $N - k$. Section 7 stress-tests departures from this model (ordered drift); our validity claims are stated relative to it, and the carrier limitation (counts only, no order) is made explicit in Section 8.

Let $MV(k) = \mathbf{1}\{x_k > k/2\}$ with ties assigned to the negative side, and let $MV(N) = \mathbf{1}\{K > N/2\}$ be the full- N *binary pass-count replay decision*. It is not an aggregation of delivered textual answers or an actual delivered answer-majority: the replay model contains only binary pass/fail outcomes. The *flip event* of a stopping rule τ is $\{MV(\tau) \neq MV(N)\}$.

Remark 1 (Why the mixture, not the correlation). On τ^2 -bench carriers (6 public task suites, 4 rollouts/task) we measured pooled within-task reward correlation $\hat{\rho} = 0.47\text{--}0.59$, but a task-demeaned (fixed-effect) estimate gives $\hat{\rho}_{FE} \approx -1/3 \approx -1/(N - 1)$, the demeaning artifact floor: essentially *all* observed correlation is across-task heterogeneity, not latent within-task dependence. On

OpenMathReasoning the per-task pass-rate variance exceeds the binomial baseline by $34\times$. The flip probability is driven by the mass of tasks with $K \approx N/2$ — a mixture functional — so pooled second-moment discounts are the wrong certificate statistic.

4 METHOD: THE MIXTURE-POSTERIOR STOPPER

Exact per-task flip probability. Given count K , the prefix count x_k is hypergeometric $\text{HG}(N, K, k)$, hence

$$f_K(k) := \mathbb{P}(\text{MV}(k) \neq \text{MV}(N) \mid K) = \sum_x \text{HG}(N, K, k, x) |\text{side}(x, k) - \text{side}(K, N)|. \quad (1)$$

Lemma 1 (Exactness and monotonicity). (1) is an identity (no approximation). For every fixed K , $f_K(k)$ is nonincreasing in odd k ; hence for any mixture H , $\mathbb{E}_H[f_K(k)]$ is nonincreasing and $k^*(\alpha) = \min\{k : \mathbb{E}_H[f_K(k)] \leq \alpha\}$ is well defined and binary-searchable.

Proof. The identity is a definition expansion. Monotonicity follows from a coupling of the prefixes at k and $k+2$ on a common ordered draw. The proof first handles the support-boundary case where the downward boundary atom has probability zero directly; only otherwise does it divide to obtain the ratio $(K - t)/(N - K - t) \geq 0$ for $K > N/2$. The complementary case follows by pass/fail reflection (Appendix A). An exact-rational CPU audit checks every admissible (N, K, k) for $2 \leq N \leq 64$, including (32, 17, 29). $\square$

Conditional certificate. Given prior H and an observed prefix (x, k) , define the posterior flip probability

$$c_H(x, k) := \mathbb{P}_H(\text{side}(K, N) \neq \text{side}(x, k) \mid x, k), \quad \text{posterior} \propto H[K] \cdot \text{HG}(N, K, k, x). \quad (2)$$

We define the terminal certificate by $c_H(x, N) := 0$. At a posterior-supported terminal state, the hypergeometric likelihood has support only on $K = x_N$, so this convention equals the conditional flip probability: $c_H(x_N, N) = 0$. It also gives the correct endpoint value for a zero-prior-mass row of a frozen certificate table, because the observed total count already fixes the binary full- N replay decision.

Theorem 1 (Adaptive is free). Let τ be any stopping time measurable w.r.t. the prefix filtration that stops only at states with $c_H(x_\tau, \tau) \leq \alpha$. The terminal state $k = N$ satisfies this same rule because $c_H(x_N, N) = 0$. Then $\mathbb{P}(\text{MV}(\tau) \neq \text{MV}(N)) = \mathbb{E}[c_H(x_\tau, \tau)] \leq \alpha$, with equality in the identity and no union bound over stopping times.

Proof. $\mathbb{P}(\text{flip}) = \mathbb{E}[\mathbb{P}(\text{flip} \mid \text{prefix path})]$. For any path stopped at (x, k) , $\mathbb{P}(\text{MV}(N) \neq \text{side}(x, k) \mid \text{path}) = c_H(x, k)$ since the posterior is exactly the conditional law of K and $\text{MV}(N)$ is a function of K . At $\tau = N$, $x_N = K$ and hence this conditional probability is $c_H(x_N, N) = 0$; at every earlier stopped state it is at most α by the rule. Thus the integrand is bounded pointwise by α , and taking expectations gives the claim. $\square$

BAYES-H is the rule $\tau = \min\{k \geq 3 : c_H(x_k, k) \leq \alpha\}$ with $H = \hat{H}_{\text{FIT}}$, the FIT-frozen empirical mixture. Its per-task exact flip probability and expected stopping time under any true count K are computable by a linear-time DP over prefix states (k, x) (Appendix B), enabling exact calibration and evaluation without Monte Carlo.

Theorem 2 (CAL population layer for a FIT-frozen rule family). Let FIT and CAL be independent i.i.d. task samples from a common count-exchangeable replay population with count law H . Conditional on FIT, freeze a finite pre-declared family $\mathcal{R}_{\text{FIT}}$ (including all certificate tables), and fix $r \in \mathcal{R}_{\text{FIT}}$. Let $g_r(K)$ be its exact per-task replay flip probability and let $\bar{g}_{\text{CAL},r}$, $s_{\text{CAL},r}^2$ be the mean and variance of $g_r(K)$ over the m CAL tasks. For a per-rule failure allocation δ_r , with probability $1 - \delta_r$ over CAL conditional on FIT,

$$\mathbb{P}_{\text{replay}, H}(\text{flip}) \leq \bar{g}_{\text{CAL},r} + \sqrt{\frac{2s_{\text{CAL},r}^2 \log(4/\delta_r)}{m}} + \frac{7 \log(4/\delta_r)}{3(m-1)},$$

where the first term is $\bar{g}_{\text{CAL},r}$. This is the empirical-Bernstein inequality (Maurer & Pontil, 2009) for bounded i.i.d. per-task flips $g_r \in [0, 1]$. Setting $\delta_r = \delta/J$ and applying Bonferroni gives a simultaneous CAL selection guarantee over J frozen candidates (Section 5).

Remark 2 (No Jensen gap). The population flip rate under replay is a *linear* functional of H : $\mathbb{P}(\text{flip}) = \sum_K H[K] g(K)$. Hence estimation error in $\hat{H}_m$ enters only through the empirical mean of a bounded per-task quantity — there is no Jensen correction, and empirical Bernstein is essentially tight at $m = 4000$.

Proposition 1 (Exact deterministic TV sensitivity of a frozen rule). *Fix the rule (certificate tables frozen from $\hat{H}$) and let $g(K)$ be its exact per-task flip probability. Over the total-variation ball $B_R = \{q : q \geq 0, \sum_K q_K = 1, \|q - \hat{H}\|_1 \leq 2R\}$,*

$$V(R) := \max_{q \in B_R} \sum_K q_K g(K)$$

is a capacity-constrained linear program whose optimum is a two-sided greedy: add mass to atoms in descending g (capped by $q_K \leq 1$) and drain mass from atoms in ascending g (capped by $\hat{H}_K$). It is computable in $O(N \log N)$ and is nondecreasing, piecewise linear in R ; it is not generally globally affine because donor and recipient capacities create breakpoints. Since B_1 is the full simplex, $V(1) = \max_K g(K)$. For the positive radii and realized capacities checked in the supplied LP grids, it is strictly below the 1-Lipschitz envelope $\text{flip}(\hat{H}) + R$; no universal strict-refinement claim is made. The critical radius $\tau^(\alpha) = \max\{R : V(R) \leq \alpha\}$ is a deterministic property of the frozen rule and fitted reference mixture. It becomes a population certificate only after separately proving that an unknown target mixture lies in B_R , which current artifacts do not do.*

On the OMR carrier the critical radii are $\tau^*(0.10/0.05/0.02/0.01) = 0.144/0.078/0.074/0.052$ (shard 0) and $0.157/0.082/0.076/0.024$ (shard 1) (`tv_robustness_r481_result.json`; the closed form matches a direct simplex LP to 10^{-6} on a $6 \times 4 \times 2$ grid of radii, levels and shards). Three consequences: (i) at the observed cross-shard mismatch $R = 0.042$, the fixed-rule sensitivity calculation returns 0.039 at the 0.05 table and 0.016 at the 0.02 table. It describes an assumed ball around a fitted mixture, not containment of either shard’s unknown population mixture; at 0.01, $\tau^*(0.01) = 0.024 < 0.042$. (ii) Tightening a frozen table gives a deterministic B_R sensitivity calculation (at $R = 0.05$, the 0.10 table re-evaluated at 0.05 has $V(R) = 0.042$ and 80.3% replay count reduction), not a “validity band” without a target-mixture confidence set; (iii) at every level a 36–47% mass of atoms has $g(K) = 0$ exactly (the certificate stops only on states the prefix already decides), which is why $V(R)$ grows far slower than the Lipschitz rate $R \cdot \max_K g(K)$. *Cross-carrier*. On the RLVE carrier ($N=8$; same frozen FIT prior as Section 6, no TEST re-read) the same closed form gives $\tau^*(0.10/0.05/0.02/0.01) = 0.199/0.551/0.213/0.449$ (`tv_robustness_rlve_r482_result.json`; LP cross-check matches to 10^{-6}). The radii are *larger* than on OMR at every level even though the RLVE prior is no more heterogeneous (het-excess variance 0.0999 vs 0.129): the RLVE prior concentrates 74–87% of its mass on *zero- g* atoms (vs 36–47% on OMR), and with only $N=8$ atoms the worst-case adversary has few high- g directions in which to move mass. Across carriers τ^* co-varies with the *alignment* of the prior with the zero- g set and is uncorrelated with heterogeneity (het-excess vs τ^* : Spearman 0.01; het-excess vs alignment: 0.05 — the two candidate drivers are not confounded). The alignment- τ^* co-variation is a *between-carrier* law: pooled Spearman 0.75 over the 16 carrier $\times$ level cells, 1.0 over the four carrier means, 0.63 with the two whole-simplex-plateau cells removed; *within* OMR it is negative (−0.95/−0.89 per shard — tightening the level moves the zero- g set only at the most-aggressive end, where mass is added on the lowest- g atoms), positive within RLVE (0.32) and OpenR1 (1.0) (decomposition audit: `spearman_decomposition_r486_result.json`; Appendix E). (Two disclosures: τ^* bisection uses $R \in [0, 1]$ here (0.551 > 0.5 exceeds the $[0, 0.5]$ range sufficient for OMR); τ^* is non-monotone in α — $\tau^*(0.05) > \tau^*(0.10)$ — because tightening α zeroes $K=6$ ($g : 0.107 \rightarrow 0$) and shrinks $K=5$ ($0.339 \rightarrow 0.071$), flattening the g -profile’s top faster than the budget tightens; internal drops exist as well (the curve has 7 rule-change breakpoints on the plotted range, Figure 2), so τ^* should not be read as globally monotone in α .) *Three-atom edge case*. On OpenR1 ($M=2$; same frozen FIT prior, no TEST re-read) the support is three atoms $p \in \{0, \frac{1}{2}, 1\}$ and $g(0) = g(1) = 0$ at every level (a decided prefix cannot be overturned by one more roll-out), so the worst case is a pure one-sided pump into the unique positive- g atom. Its capacity-aware curve is $V(R) = g_{\max} \min\{1, \hat{H}(\frac{1}{2}) + R\}$; on the reported sub-capacity crossings this gives $\tau^*(\alpha) = \alpha/g_{\max} - \hat{H}(\frac{1}{2})$ (`tv_robustness_openr1_r483_result.json`; scan match 10^{-9} , LP 10^{-6}). Since $g_{\max}$ tracks the certificates (0.0785/0.0460): $\tau^*(0.10) = \tau^*(0.05) = 0.168$ (halving α past 0.0785 halves $g_{\max}$, $0.25 \rightarrow 0.125$), while $\tau^*(0.02) = \tau^*(0.01) = 1$ — below the smallest certificate the frozen rule never stops, $V(R) \equiv 0$, no prior shift can break it. Non-

monotonicity is thus maximal here ($0.168 \rightarrow 1$); zero- g mass 76.8% sits between OMR’s 36–47% and RLVE’s 74–87%, completing the pattern from the low-support side. *Conservation curve.* Appendix E maps $\tau^*(\alpha)$ on the dense continuum $\alpha \in [10^{-3}, 0.2]$ for all three carriers: the curve is analytically staircase-shaped — non-decreasing inside each constant-rule interval, so that every jump of τ^* is a strict *drop* exactly at a certificate threshold (analytic on OpenR1: τ^* drops by 0.314 at $\alpha = 0.0785$ and by 0.864 at 0.0460). The practical reading is a staircase interpretation of the certificate level: at a rule-change breakpoint, a small budget relaxation buys a discrete jump in certified drift tolerance, so α should be set on a step, not inside one.

Remark 3 (Continuous- α selection is free for BAYES-H). The stopping condition $c_{\hat{H}}(x, k) \leq \alpha$ does not involve α on the left-hand side, so the BAYES-H rule family is *nested* in α : as α varies continuously, the rule changes only at the finitely many breakpoints given by the distinct values of $c_{\hat{H}}(x, k)$ — at most $\sum_{k=3}^N (k+1)$ stoppable states (555 at $N=32$; 267 distinct values observed, plus the never-stop rule). These are three distinct counts: 555 stoppable states bound the possible rule changes; only 267 *distinct* certificate values are observed on the support, and the rule family has one rule per value *plus* the never-stop rule (268 in total; shrunk to 182 below for the tested range). A union bound over this breakpoint family therefore certifies the *entire continuum* $\alpha \in (0, 0.10]$ simultaneously — no pre-declared α grid is needed. The family can be shrunk further with no loss: for the range $\alpha \in (0, 0.10]$ only the 181 distinct positive certificate values ≤ 0.10 are rule-distinguishing (values > 0.10 never trigger any rule in the range; zero-certificate states stop under every $\alpha > 0$), so the exact effective family has 182 rules ($J = 6,501$ including the FIXED- k part) instead of the 9,339 obtained by counting all observed breakpoints; the four reference selections below are unchanged and their UCBs tighten by $\leq 6 \times 10^{-4}$ (prop_562bound_r479_result.json). On our calibration pool the enlarged family ($J_{\text{cont}} = 9,339$ vs. $J = 64$) inflates UCBs only through the logarithmic factor; the selected replay-count-optimal BAYES-H rule is bit-for-bit identical at $\alpha \in \{0.10, 0.05, 0.02\}$ and tightens only at $\alpha = 0.01$ (mean k $10.3 \rightarrow 11.9$), while the weaker FIXED-EB baseline loses its $\alpha = 0.02$ budget under *either* family size (alpha_continuous_r478_result.json).

5 PROTOCOL: FIT/CAL/TEST WITH A LEGITIMATE SELECTION FAMILY

Split and status. 11,607 problems are partitioned once (seeded) into FIT (4000), CAL (4000), TEST (3607). The population statement in Theorem 2 assumes tasks are i.i.d. draws from one replay population and that FIT and CAL are independent. FIT estimates $\hat{H}$ and freezes the BAYES-H certificate tables and the pre-declared family before CAL is inspected. CAL selects rules from that frozen family: fixed- k candidates $k \in \{3, 5, \dots, 31\} \times \alpha \in \{0.10, 0.05, 0.02, 0.01\}$ (60 certificates) and BAYES-H at the same 4 levels, with simultaneous empirical-Bernstein UCBs at per-test $\delta = 0.05/64$. $k^*(\alpha)$ is the smallest budget whose CAL UCB is $\leq \alpha$; BAYES-H is accepted at level α iff its CAL UCB is $\leq \alpha$. TEST is read exactly once, is not used for selection, and is a fixed-benchmark descriptive readout rather than a second population guarantee. All evaluation uses the exact per-task flip functions ((1) for fixed- k ; the DP of Appendix B for BAYES-H), so TEST numbers are exact replay quantities, not MC estimates; the DP itself is cross-checked against replay MC (max deviation 0.043 within $3\sigma_{\text{MC}} = 0.075$).

Baselines. FULL- N ; FIXED-HOEF (fixed budget selected with distribution-free Hoeffding UCBs, the strongest model-free competitor); FIXED-EB (fixed budget with the same mixture-aware empirical-Bernstein machinery as BAYES-H — isolates the value of adaptivity); BAYES-UNIF (uniform prior; not a certificate when the prior is wrong); WINDOW3 (ESC-style: stop when 3 consecutive samples agree with the current majority — no certificate knob).

6 EXPERIMENTS

Carrier. OpenMathReasoning CoT shard 0 (Moshkov et al., 2025) (CC-BY-4.0): the pass_rate_72b_tir column gives, for 11,607 unique problems, the exact pass count K of Qwen2.5-Math-72B-TIR over $N = 32$ rollouts — no answer-extraction noise. Pass rates are strongly bimodal: 45% of problems have $p \leq 0.1$ or $p \geq 0.9$; 11.6% sit in $[0.4, 0.6]$. This carrier is a count-only carrier used for the *count-exchangeable replay* analysis: it supplies per-task counts but no rollout order, so the random-prefix assumption is a modeling condition rather than an observed chronology (Section 7).

Table 1: Fixed-benchmark descriptive TEST readout ($n = 3607$ problems, $N = 32$). “CAL-selected” means that the FIT-frozen candidate passed its CAL UCB selection criterion; it is not a TEST guarantee. “No CAL certificate” marks a reference or heuristic. Replay count reduction $= 1 - \bar{k}/32$. BAYES-H attains the largest valid replay count reduction at $\alpha = 0.05$ and 0.02 ; at $\alpha = 0.10$ FIXED-EB has 84.4% replay count reduction versus BAYES-H 84.0% (a 0.4-point gap in mean replay count on this test pool, descriptive, no paired CI computed for this pair), while BAYES-H keeps the lowest realized flip at every level. WINDOW3 has the largest replay count reduction but is invalid at $\alpha \leq 0.05$.

Rule	α	replay full-vote flip	mean k	count reduction
<i>Status. CAL-selected rows passed a FIT-frozen CAL UCB; TEST values are descriptive. FULL is a reference; WINDOW3 has no CAL certificate.</i>				
FULL-32	—	0	32.0	0
WINDOW3 (ESC-style)	—	0.0577	4.4	86.2%
FIXED-HOEF $k^*=7$	0.10	0.0645	7.0	78.1%
FIXED-EB $k^*=5$	0.10	0.0794	5.0	84.4%
BAYES-H	0.10	0.0370	5.1	84.0%
FIXED-HOEF $k^*=27$	0.05	0.0156	27.0	15.6%
FIXED-EB $k^*=17$	0.05	0.0324	17.0	46.9%
BAYES-H	0.05	0.0249	6.1	80.9%
FIXED-HOEF	0.02	<i>no certifiable budget</i>		
FIXED-EB $k^*=31$	0.02	0.0083	31.0	3.1%
BAYES-H	0.02	0.0091	8.5	73.5%

Main table. Table 1 reports the single TEST readout (Bonferroni CI radius 0.0286 across the 9 reported rules). In this section, “valid” means replay full-vote flip $\leq \alpha$; TEST is descriptive (selection was certified on CAL). Neither flip nor count reduction is gold correctness or an operational token, latency, or cancellation measurement.

Fair comparison. Against the strongest distribution-free baseline (FIXED-HOEF) at the same α , BAYES-H’s paired per-problem replay-count-reduction gap is $+0.058$ (CI radius 0.029) at $\alpha = 0.10$ and $+0.652 \pm 0.029$ at $\alpha = 0.05$ — a $5.2\times$ improvement over the baseline’s 15.6% count reduction; at $\alpha = 0.02$ the baseline has *no* certifiable budget while BAYES-H still has a 73.5% reduction. Against the strongest mixture-aware fixed budget (FIXED-EB), adaptivity adds +34 points of replay count reduction at $\alpha = 0.05$ (80.9% vs 46.9%) and +70 points at $\alpha = 0.02$ — the adaptive advantage grows as α tightens, because the conditional certificate stops early exactly on the bimodal mass (extreme- K problems stop at $k \approx 5$) and waits only on the 11.6% of mid-difficulty problems.

Cross-source transfer. Splitting CAL/TEST across the four AoPS problem sources (train on three, test on the fourth), BAYES-H stays valid on all transfers (replay flip $0.021\text{--}0.033 < \alpha = 0.05$; count reduction 79–83%). This supplied count-replay readout is descriptive of these source splits; it does not establish ordered online transfer.

Second-shard replication and prior-mismatch transfer. We replicate the entire chain on a disjoint carrier, OpenMathReasoning CoT shard 1 (11,454 problems, same $N=32$ pass counts; SHA-256 pinned), whose mixture structure is near-identical (heterogeneity excess 0.125 vs 0.129; 44.4% extreme- p mass). Two runs share the CAL/TEST problems of shard 1 and differ only in the prior’s provenance. (i) *Within-shard*: the prior is fit on shard 1’s own FIT split; every qualitative replay conclusion reproduces — BAYES-H has 80.6% replay count reduction at realized flip 0.025 ($\alpha = 0.05$) and 73.1% at 0.009 ($\alpha = 0.02$), vs 15.6% and *no certifiable budget* for FIXED-HOEF. (ii) *Cross-shard prior transfer*: the prior is fit on shard 0 FIT (fitted-mixture TV distance to the shard 1 fit $= 0.042$) and the certificate is selected and read on shard 1. All numbers move by ≤ 0.4 points (count reduction 80.6%/73.2%; flips 0.025/0.009). This is a small movement in one supplied count-replay transfer and a deterministic fitted-mixture sensitivity diagnostic, not a statistical population-transfer certificate or a demonstration of robustness to naturally ordered online rollouts (Section 3).

Second model-carrier pair: OpenR1-Math-220k (DeepSeek-R1, $M = 2$). To probe the mechanism under a different model and a structurally opposite carrier, we run the identical chain on OpenR1-Math-220k (MIT; 9,374 unique problems, two deduplicated DeepSeek-R1 rollouts each, HF snapshot e4e141ec). Under the replay model, the full vote is the 2-vote majority (ties broken by a pre-drawn fair coin), the only stoppable prefix is $k = 1$, and per-problem risk is analytic. The only identifiable prior at $M = 2$ is coarse (three atoms $p \in \{0, \frac{1}{2}, 1\}$, fit by exact pair-type moment matching): 25% always-fail, 23% coin-flip, 52% always-pass. Results (FIT/CAL/TEST 3000/3000/2853): (i) the conditional certificate attains the replay risk-count *frontier*: at $\alpha = 0.20$ it stops on every problem (flip $0.059 \leq \alpha$, 50% replay count reduction); at $\alpha = 0.10$ full stopping has the tightest distribution-free UCB $0.080 \leq \alpha$ and is certified with the same realized flip and count reduction; at $\alpha = 0.05$ it stops only on the sub-population whose observed prefix certifies (positive-side prefix), giving 32% replay count reduction at flip $0.029 \leq \alpha$; at $\alpha = 0.02$ it *never* stops — and no legal certificate can, since stopping at $k=1$ on any problem carries posterior risk > 0.02 ; (ii) the realized flip of always stopping at $k=1$ (0.059) matches its analytic value $\mathbb{E}_H[p(1-p)] = 0.058$ to Monte-Carlo precision. Replay count reduction is bought by mixture *structure* that a prefix can resolve: with $M = 2$ there is at most one draw of prefix information, and the certificate extracts exactly the count reduction that structure allows — 50% at loose levels, partial at $\alpha = 0.05$, honestly nothing at $\alpha = 0.02$ — versus 81%/73% at $\alpha = 0.05/0.02$ on the $N=32$ OMR carrier. This is cross-model replay evidence for the stated mechanism; it does not test chronological online deployment, gold correctness, or operational cost.

Third model-carrier pair: RLVE (Qwen3-4B, $N = 8$). The two carriers above close the extremes ($N=32$ rich-prefix vs $M=2$ near-prefix-free). To probe the intermediate regime under a third model family and a different task family, we run the identical chain on RLVE teacher rollouts (Qwen3-4B-Thinking-2507, pass@8 on 9,000 verifiable counting/combinatorics environments; Apache-2.0; HF snapshot eaeec946). Success is the offline Gym-verifier reward being positive (a small mass of fractional partial credit is counted as failure). The mixture is again strongly U-shaped (42% extreme problems, heterogeneity excess 0.100 over the binomial baseline). Results (FIT/CAL/TEST 3000/3000/3000): (i) BAYES-H has replay count reduction 57.6%/50.9%/48.2%/45.4% at $\alpha = 0.10/0.05/0.02/0.01$ with realized flips 0.031/0.010/0.005/0.002 — conservative at every level; (ii) the certificate passes at *every* level down to $\alpha = 0.01$, while *no* distribution-free budget is certifiable below $\alpha = 0.10$ (FIXED-EB and FIXED-Hoeffding both null at $\alpha \leq 0.05$) — at $\alpha = 0.10$ the paired replay-count-reduction gap over FIXED-Hoeffding is $+0.20$ (CI radius 0.03, significant); (iii) the exact DP prediction matches a 3,000-draw replay Monte-Carlo on held-out problems (flip within 0.004, mean k within 0.04). Two descriptive diagnostics: the recorded sample-id success slope is flat (pooled slope -7.8×10^{-5} /trial; per-position rates 0.316–0.326), consistent with near-exchangeable sampling in this carrier; and the replay count reduction interpolates with resolvable structure exactly as the mechanism predicts (81% at $N=32$ with 15 stoppable budgets, 45–58% at $N=8$ with 3, 32–50% at $M=2$ with 1; all in-distribution replay readouts collected in Appendix F). The sample-id diagnostic does not audit an online policy or make the count-replay certificate valid for deployment order.

Why the window heuristic fails. On mid-difficulty problems ($K \approx 16$), three consecutive agreements with the current majority is a *common* event under near-coin-flip draws, not evidence the vote has settled: WINDOW3 realizes flip 0.058 at mean $k = 4.4$. Its flip rate is nearly flat in α (it has no certificate knob): it can only be tuned by window length, and its certified population UCB is 0.061 (adaptive_sweep_r468.result.json), so it barely misses even the loosest tested level $\alpha = 0.10$ on the full OMR test pool.

7 SYNTHETIC ORDERED DRIFT: FAILURE DOMAIN AND MARGIN REPAIR

The replay model assumes prefixes are exchangeable with the held remainder. Because the carrier contains no observed chronology, we apply three *pre-fixed synthetic* drift mechanisms to its true counts ($\delta \in \{0, 0.05, 0.10, 0.15, 0.20\}$, 2000 TEST problems, 500 orders each): **E1** (front-load: the first k draws are enriched/depleted in passes by count $D = \lceil 2\delta \min(K, N-K) \rceil$), **E2** (linear per-draw pass-rate trend $p_j = K/N + \delta(2j/(N-1)-1)$), **E3** (adversarial block: the last $\lceil \delta N \rceil$ draws drift toward the minority side).

Findings (full tables in Appendix D): (i) all methods break under E1 by $\delta = 0.10$ – 0.15 in this harness; the certificate is honest about its premise — E1 violates exchangeability by construction; (ii) under the gentler E2/E3, BAYES-H is the *most* robust stopper in this synthetic grid: at $\alpha = 0.10$ it remains valid to $\delta = 0.20$ under E2 (flip 0.083) while FIXED- k breaks at $\delta = 0.15$ and WINDOW3 breaks at $\delta = 0.20$ (flip 0.104; under E3 WINDOW3 stays valid throughout, its last-window rule never reaching the drifting tail); at $\alpha = 0.05$ under E3 it breaks only at $\delta = 0.15$ (flip 0.052), while FIXED-EB survives (it stops at $k = 17$, having seen half the block); (iii) an exploratory margin scan illustrates a synthetic flip/count tradeoff. No frozen margin-selection record is bundled, so its thresholds are not confirmatory. Under E3 at $\alpha = 0.05$, the explored margin $\gamma = 0.025$ covers $\delta \leq 0.15$ (flip 0.0495, replay count reduction 77.2% vs 81.0% unmarginated in the stress harness; the unmarginated exact TEST readout of Table 1 is 80.9%); covering the extreme $\delta = 0.20$ adversarial block requires $\gamma > 0.03$ (count-reduction cost > 7 points), and under the smoother E2 trend $\gamma = 0.03$ covers the full tested range $\delta \leq 0.20$ (flip 0.0491, count reduction 71.8%). The drift budget δ could be estimated on CAL only if ordered rollouts are retained; on this count-only carrier it is assumed. This synthetic analysis neither measures nor certifies natural online order.

8 LIMITATIONS

(i) The public carrier provides pass *counts*, not ordered rollouts: all main results are exact only under the count-exchangeable replay model; the synthetic stress mechanisms of Section 7 only characterize specified departures from that model. The certificate is a replay-model conditional certificate, not an e-process: validity under arbitrary online stopping times in ordered deployment is outside the stated claims. No supplied run compares the stopped answer with gold correctness, or records generated tokens, latency, cancellation acknowledgement, or post-cancellation work; full-vote agreement and $1 - \bar{k}/N$ cannot substitute for those endpoints. (ii) The mixture prior is estimated from 4000 calibration problems of the same distribution; adversarial mixture shift beyond the tested transfer range can break the certificate; the exact critical radii of Proposition 1 give deterministic frozen-rule sensitivity radii over an assumed fitted-mixture TV ball ($\tau^* = 0.024$ – 0.157 across levels and shards); they do not establish target-population containment. Within that assumed ball the failure boundary is sharp — at $\alpha = 0.01$ on shard 1 the observed cross-shard mismatch 0.042 already exceeds $\tau^*(0.01) = 0.024$; and the radii are *rule-conditional*: $\tau^*(\alpha)$ is a staircase with jumps at the certificate thresholds (Appendix E), so the level α should be read as choosing a step of that staircase, not a point of a smooth trade-off. Along the estimation axis, Appendix E(e) shows τ^* is not monotone in the prior-fit size m ; the decomposition of Appendix E(f) attributes this to the *rule* the fit induces (freezing the prior leaves the non-monotonicity intact; freezing the rule removes the largest dips), not the prior’s sampling error. Appendix E(g) now records only an *oracle K-wise profile-capping sensitivity*: its stored subset is selected separately after conditioning on the final count K . Because that final count is unavailable at a prefix, this diagnostic is neither a common prefix-state rule nor an instance of Theorem 1; it supports no stopping-policy, cost, or robustness-repair claim. Appendix F collects every in-distribution readout and its failure boundary in one matrix. (iii) Lemma 1’s monotonicity is proved in closed form (Appendix A) and independently verified exhaustively at $N = 32$. (iv) Main-table counts come from one model–task pair (Qwen2.5-Math-72B-TIR, math problems); generality across problem pools is supported by the disjoint second-shard replication and cross-shard prior transfer, and generality across models and task families by two further model–carrier pairs — DeepSeek-R1 rollouts on OpenR1-Math-220k ($M = 2$), on which the certificate attains its replay risk–count frontier, and Qwen3-4B rollouts on RLVE verifiable environments ($N = 8$), on which it is certified at every tested level and interpolates with the resolvable mixture structure (Section 6). (v) At $k = N$, the observed pass count equals K , so the binary pass-count replay decision agrees with itself and $c_H(x_N, N) = 0$. Terminal stops add zero replay flip for every true count; this endpoint convention is shared by Theorem 1 and the DP in Appendix B.

9 CONCLUSION

Treating across-task heterogeneity as the certificate object turns early stopping for majority voting into an exact conditional-inference problem under the replay model: adaptivity is free (tower property), calibration is one empirical-Bernstein step over problems, and the supplied OMR readout has an 81% replay rollout-count reduction at full-vote flip level 0.05, where the strongest distribution-

free baseline has 16% and the strongest fixed mixture-aware budget 47%. The same framework admits a synthetic ordered-drift analysis with a margin repair, while natural online order, gold correctness, and actual token/latency/cancellation costs remain open.

REPRODUCIBILITY STATEMENT

The count-replay analyses, assumptions, split protocol, and complete proofs are specified in Sections 3–5 and the appendix. The supplied artifacts, pinned carrier checksums, deterministic seeds, and claim checker are indexed in the anonymous supplementary reproduction package; its current limitations, including unavailable upstream carrier paths and the absence of ordered online telemetry, are stated in Section 8.

AI-USE STATEMENT

Generative AI tools were used to assist manuscript-language revision, evidence and citation auditing, build checks, and internal pre-submission review, including the Codex multi-agent revision and audit documented for this version. A generative image tool was also used to compose Figure 1’s conceptual layout; its quantitative labels were independently checked against the cited on-disk result artifacts. These tools did not generate, execute, or replace the reported scientific experiments, their quantitative result artifacts or values, or the final technical judgments; the authors reviewed the changes and remain responsible for the paper.

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A PROOFS

A.1 MONOTONICITY OF THE FIXED- k FLIP PROBABILITY (LEMMA 1)

Recall

$$f_K(k) = \mathbb{P}(\text{MV}(k) \neq \text{MV}(N) \mid K), \quad x_k \sim \text{HG}(N, K, k),$$

with the negative side winning ties ($\text{MV}(k) = 1$ iff $x_k > k/2$). We prove that $f_K(k)$ is non-increasing along the odd numbers $k \in \{1, 3, 5, \dots, N-1\}$ for every fixed K ; this suffices for the paper, since the fixed- k certificate family is evaluated on odd budgets only, and it makes $k^*(\alpha) = \min\{k : \mathbb{E}_H[f_K(k)] \leq \alpha\}$ well defined and binary-searchable for every mixture H (mixtures of nonincreasing functions are nonincreasing).

Setup and case split. $\text{MV}(k) \neq \text{MV}(N)$ can happen in only two ways:

- **Case U (upset).** $K \leq N/2$ (full vote negative) and $x_k > k/2$ (prefix votes positive).
- **Case R (reversal).** $K > N/2$ (full vote positive) and $x_k \leq k/2$ (prefix votes negative).

Write $f_K(k) = U_K(k) + R_K(k)$ with $U_K(k) = \mathbb{P}(x_k > k/2) \mathbf{1}\{K \leq N/2\}$ and $R_K(k) = \mathbb{P}(x_k \leq k/2) \mathbf{1}\{K > N/2\}$. At most one of the two terms is nonzero for any K .

The reflection identity. Swapping the labels “pass”/“fail” of all N rollouts maps the law $\text{HG}(N, K, k)$ of x_k onto the law of $k - x_k$ under count $N - K$, i.e. $\mathbb{P}_K(x_k > s) = \mathbb{P}_{N-K}(x_k < k - s)$. Hence, for odd k ,

$$U_K(k) = \mathbb{P}_K(x_k > k/2) = \mathbb{P}_{N-K}(x_k < k/2) = R_{N-K}(k) \quad (K \leq N/2),$$

since k odd makes $k/2$ a half-integer, so the strict and weak inequalities coincide. Every upset probability thus equals a reversal probability at the reflected count, and it suffices to prove

$$R_K(k+2) \leq R_K(k) \quad \text{for all } K > N/2 \text{ and odd } k, \quad (3)$$

which covers Case R directly and Case U by reflection (the same display applied at count $N - K$).

Coupling construction. Fix $K > N/2$ and odd $k \leq N - 3$. We couple the two hypergeometric draws on a single probability space. Let the population of N rollouts consist of K passes and $N - K$ fails. Draw an *ordered* sample of size $k + 2$ uniformly without replacement and split it as

$$\underbrace{X_1, \dots, X_k}_{\text{prefix } k} \mid \underbrace{X_{k+1}, X_{k+2}}_{\text{extra pair } (A, B)}.$$

The marginal law of $(X_1, \dots, X_k)$ is exactly $\text{HG}(N, K, k)$, and $x_{k+2} = x_k + A + B$ has law $\text{HG}(N, K, k+2)$. All statements below are about this joint space, so they hold for the two marginals simultaneously.

Let $t = k/2 + 1/2$ be the (integer) flip threshold: the prefix at k votes negative iff $x_k \leq t - 1$; the prefix at $k + 2$ votes negative iff $x_{k+2} \leq t$. Partition the event $\{x_k = s\}$ by the value of the pair $A + B \in \{0, 1, 2\}$:

$$\begin{aligned} E_0 &= \{x_k = t - 1, A + B = 0\}, & \text{flip at } k, & \text{flip at } k + 2, \\ E_1 &= \{x_k = t - 1, A + B = 1\}, & \text{flip at } k, & \text{flip at } k + 2, \\ E_2 &= \{x_k = t - 1, A + B = 2\}, & \text{flip at } k, & \text{no flip at } k + 2, \\ F_0 &= \{x_k = t, A + B = 0\}, & \text{no flip at } k, & \text{flip at } k + 2, \\ F_1 &= \{x_k = t, A + B = 1\}, & \text{no flip at } k, & \text{no flip at } k + 2. \end{aligned}$$

All other states agree on both sides (e.g. $x_k \leq t - 2$ flips under both, since $A + B \leq 2$; $x_k \geq t + 1$ flips under neither). Therefore

$$R_K(k) - R_K(k+2) = \mathbb{P}(E_2) - \mathbb{P}(F_0), \quad (4)$$

and it remains to show $\mathbb{P}(E_2) \geq \mathbb{P}(F_0)$.

Comparison of the two boundary atoms, including support boundaries. Both atoms factor as (prefix probability) $\times$ (pair probability). We first split off the case $\mathbb{P}(F_0) = 0$. Then (4) gives $R_K(k) - R_K(k+2) = \mathbb{P}(E_2) \geq 0$ directly; in particular, no ratio is formed when a binomial factor in F_0 vanishes (including a zero remaining-fail count or an infeasible $x_k = t$ atom). It remains to consider $\mathbb{P}(F_0) > 0$. This support condition makes every denominator below strictly positive and makes the ratio algebra legitimate. With $k = 2t - 1$, first,

$$\mathbb{P}(x_k = t - 1) = \frac{\binom{K}{t-1} \binom{N-K}{t}}{\binom{N}{k}}, \quad \mathbb{P}(x_k = t) = \frac{\binom{K}{t} \binom{N-K}{t-1}}{\binom{N}{k}},$$

since $k - (t - 1) = t$ and $k - t = t - 1$. Given $x_k = t - 1$, the remainder holds $K - t + 1$ passes out of $N - k$, so $\mathbb{P}(A + B = 2 \mid x_k = t - 1) = \binom{K-t+1}{2} / \binom{N-k}{2}$; given $x_k = t$, the remainder holds $N - K - t + 1$ fails, so $\mathbb{P}(A + B = 0 \mid x_k = t) = \binom{N-K-t+1}{2} / \binom{N-k}{2}$. The denominators $\binom{N}{k} \binom{N-k}{2}$ are common to both atoms. Under $\mathbb{P}(F_0) > 0$, hence

$$\frac{\mathbb{P}(E_2)}{\mathbb{P}(F_0)} = \frac{\binom{K}{t-1} \binom{N-K}{t} \binom{K-t+1}{2}}{\binom{K}{t} \binom{N-K}{t-1} \binom{N-K-t+1}{2}} = \frac{K - t}{N - K - t},$$

where the last equality uses the three elementary ratios $\binom{K}{t-1} / \binom{K}{t} = \frac{t}{K-t+1}$, $\binom{N-K}{t} / \binom{N-K}{t-1} = \frac{N-K-t+1}{t}$, and $\binom{K-t+1}{2} / \binom{N-K-t+1}{2} = \frac{(K-t+1)(K-t)}{(N-K-t+1)(N-K-t)}$, whose product telescopes to the displayed ratio. Therefore, with $C := \binom{K}{t} \binom{N-K}{t-1} \binom{N-K-t+1}{2} / [\binom{N}{k} \binom{N-k}{2}] \geq 0$,

$$\mathbb{P}(E_2) - \mathbb{P}(F_0) = C[(K - t) - (N - K - t)] = C(2K - N) \geq 0, \quad (5)$$

because $K > N/2$ in Case R. The excluded support case was already proved directly; if $t > K$, $\mathbb{P}(x_k = t - 1) = 0$ and both atoms vanish. By (4) this proves (3) and, by the reflection identity, $U_K(k+2) \leq U_K(k)$ as well, completing the proof of monotonicity along odd k . $\square$

Remark 4 (Verification at $N = 32$). Independently of the proof, the revision's exact-rational CPU audit enumerates all admissible (N, K, k) for $2 \leq N \leq 64$, including the boundary $(N, K, k) = (32, 17, 29)$, checks (4), the support split, and the monotonicity inequality. The historical $N = 32$ artifact remains a separate frozen result check.

Remark 5 (Why only odd k enters the certificate). Requiring k odd removes tie-breaking from the certificate family (every prefix has a strict majority side), and the binary search for $k^*(\alpha)$ is over the odd ladder only. Nothing else in the paper uses monotonicity at even k .

A.2 TERMINAL-STOP BOOKKEEPING (THEOREM 1)

The stopping rule is $\tau = \min\{k \in \{3, \dots, N\} : c_H(x_k, k) \leq \alpha\}$. At a reachable terminal state, all N binary outcomes have been observed, hence $x_N = K$. We define the endpoint certificate to be zero; whenever the posterior is supported, this also follows because the hypergeometric likelihood at (x_N, N) has support only on $K = x_N$:

$$c_H(x_N, N) = 0.$$

Thus $k = N$ satisfies the same threshold rule rather than being an exceptional terminal state. The tower argument applies to this bounded stopping time, and

$$\mathbb{P}(\text{flip}) = \mathbb{E}[c_H(x_\tau, \tau)] = \mathbb{E}[c_H \mathbf{1}\{\tau < N\}] + \mathbb{E}[c_H \mathbf{1}\{\tau = N\}].$$

The first term is at most $\alpha \mathbb{P}(\tau < N)$, and the second term is zero. Therefore the integrand is bounded pointwise by α and $\mathbb{P}(\text{flip}) \leq \alpha$. For each fixed true count K , a path reaching N also has $x_N = K$, so its terminal contribution to the DP flip probability $g(K; \alpha, \hat{H}_m)$ is exactly zero. The population layer remains needed for calibration-prior estimation, not to absorb a terminal flip.

Remark 6 (The terminal-stop mass is small in practice). On the TEST split, the exact fraction of paths reaching $k = N$ under BAYES-H is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$ (the certificate stops early exactly on the bimodal mass). Their contribution to replay flip is zero; their contribution to the expected stopping count is included exactly (Appendix B).

A.3 LINEARITY OF THE POPULATION FLIP RATE

For the population statement, FIT and CAL are assumed independent i.i.d. task samples from the same count-exchangeable replay population. Conditional on FIT, every prior, certificate table, and candidate rule is frozen before CAL is inspected. For such a fixed rule and fixed prior $\hat{H}_{\text{FIT}}$ used inside the certificate, the per-task flip probability $g(K)$ depends only on the true count K (via the hypergeometric law of prefixes) and on $\hat{H}_{\text{FIT}}$ (through the posterior threshold). The CAL values $g(K_i)$ are consequently i.i.d. bounded observations conditional on FIT, and the population replay flip rate is

$$\mathbb{P}_{\text{replay}, H}(\text{flip}) = \sum_{K=0}^N H[K] g(K),$$

a linear functional of the true mixture H . Consequences used in the main text: (i) no Jensen gap — conditional on FIT, empirical Bernstein on the CAL mean bounds the population quantity directly for a frozen rule, with each per-task quantity $g \in [0, 1]$; (ii) graceful degradation — if the deployment mixture shifts from H to H' , $|\mathbb{P}_{H'} - \mathbb{P}_H| \leq \|H' - H\|_1$ since $g \in [0, 1]$; (iii) Bonferroni over the pre-declared family of $J = 64$ rules legitimizes CAL-side min- k selection, because simultaneous UCBs on all J frozen linear functionals imply the UCB of the CAL-selected rule. The fixed benchmark TEST split is not used in this argument and remains descriptive.

B EXACT DYNAMIC PROGRAM FOR THE ADAPTIVE RULE

We compute, for a fixed true count K and frozen prior $\hat{H} = \hat{H}_m$, the exact per-task flip probability $g(K; \alpha, \hat{H})$ and expected stopping time $\mathbb{E}[\tau \mid K]$ of BAYES-H under the replay model, by dynamic programming over prefix states.

State space and transitions. A state is a prefix (k, x) with $0 \leq x \leq k \leq N$, $k \geq 0$. Under count K , the next draw is a uniform element of the remainder, so the transition kernel is

$$\mathbb{P}(x_{k+1} = x + 1 \mid k, x, K) = \frac{K - x}{N - k}, \quad \mathbb{P}(x_{k+1} = x \mid k, x, K) = \frac{N - K - (k - x)}{N - k},$$

and the initial state has $\mathbb{P}(x_0 = 0) = 1$. The state graph is a DAG ($O(N^2)$ states, $O(N^2)$ edges), so a single backward pass is linear in the number of states.

Certificate table. The posterior $\mathbb{P}(K = j \mid k, x) \propto \hat{H}[j] \text{HG}(N, j, k, x)$ and $c_H(x, k)$ are pre-computed once for all (k, x) on the $O(N^2)$ grid (with the terminal convention $c_H(x, N) := 0$). At a posterior-supported endpoint, the likelihood supports only $j = x$, so this convention is the conditional flip probability; it also defines a zero-prior-mass table row, where observing all outcomes still fixes the full- N binary pass-count decision. A state is a *stop state* iff $k \geq 3$ and $c_H(x, k) \leq \alpha$; therefore $k = N$ satisfies the same stopping test, rather than a separate truncation rule.

Recursions. For a fixed K , define $V(k, x) = \mathbb{P}(\text{MV}(\tau) \neq \text{MV}(N) \mid k, x, K)$ and $T(k, x) = \mathbb{E}[\tau - k \mid k, x, K]$. Then

$$V(k, x) = \begin{cases} 1\{\text{side}(x, k) \neq \text{side}(K, N)\}, & (k, x) \text{ stop state,} \\ \frac{K - x}{N - k} V(k + 1, x + 1) + \frac{N - K - k + x}{N - k} V(k + 1, x), & \text{otherwise,} \end{cases}$$

and the same recursion with $T(\cdot) = 0$ at stop states and $T(k, x) = 1 + (\text{continuation mean})$ otherwise. The requested quantities are $g(K) = V(0, 0)$ and $\mathbb{E}[\tau \mid K] = T(0, 0)$. The per-task exact saving is $1 - \mathbb{E}[\tau \mid K]/N$; population quantities follow by linearity, $\sum_K H[K]g(K)$, for any mixture H . The same forward pass yields the exact terminal-stop occupancy $\mathbb{P}(\tau = N \mid K)$. At that boundary the reachable state has $x = K$, so the stop-state value in the recursion is zero for replay flip; on the TEST split terminal occupancy is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$.

Complexity. For each K : $O(N^2)$ states, $O(1)$ work per state after the certificate table is frozen, so all 33 values of K cost $O(N^3) \approx 3.6 \times 10^4$ flops at $N = 32$ — negligible. This is why every TEST number in the main table is an *exact replay quantity*, not a Monte Carlo estimate.

Cross-check against simulation. The DP is validated against an independent replay Monte Carlo implementation (fresh random prefixes) on the same frozen prior: the maximum absolute deviation over the checked cells is 0.043, within the pre-set tolerance $3\sigma_{MC} = 0.075$ (the check includes cells at $\alpha = 0.01$, whose realized flips are small and inflate the pooled MC standard error); PASS. See `fit_cal_test_r469.py` and the self-check block of `fit_cal_test_r469_result.json`.

C FEATURE COMPARISON AND APPENDIX ROADMAP

This appendix separates proof (Appendix A), exact DP bookkeeping (Appendix B), synthetic diagnostics, and fixed-rule sensitivity. None expands the main count-exchangeable replay claim to online correctness or operational cost.

Work	Endpoint	Order/model	Guarantee / selection
ASC (Aggarwal et al., 2023); ESC (Li et al., 2024)	adaptive self-consistency	implementation-specific sequential sampling	no claim here about their universal full-vote certificate
MARS (Chen et al., 2026)	parallel trace vote movement	parallel traces/checkpoints	conservative future-movement bound; distinct from this replay posterior
Conformal (Vovk et al., 2005); CS (Howard et al., 2021)	coverage / time-uniform mean bounds	exchangeable / i.i.d.-sequential	established marginal or anytime tools, not this conditional flip object
This work	full- N binary pass-count replay disagreement	count-exchangeable domain prefix	posterior flip certificate; FIT/CAL rule selection, TEST replay readout

Table 2: Source-verified feature comparison. Relationships are deliberately endpoint- and model-specific; this table makes no priority or ordered-online claim.

D FULL DRIFT TABLES

Tables 3 and 4 report the complete drift stress grid behind Section 7 (`drift_stress_r469_result.json`; 2000 TEST problems, 500 orders per cell). Under the smooth mechanisms E2/E3, BAYES-H is the most robust stopper at both levels; under the premise-breaking E1 front-load all certificates fail together, as they should.

Table 3: Full drift stress table at $\alpha = 0.05$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold:** realized flip exceeds α (certificate broken).

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	2.5 (6.1)	3.2 (17.0)	1.6 (27.0)	5.8 (4.4)
	0.05	4.8 (6.1)	4.9 (17.0)	4.1 (27.0)	7.1 (4.4)
	0.1	7.6 (6.2)	8.2 (17.0)	7.6 (27.0)	9.6 (4.3)
	0.15	11.4 (6.0)	12.1 (17.0)	12.4 (27.0)	12.2 (4.2)
	0.2	13.0 (5.5)	13.7 (17.0)	13.6 (27.0)	13.5 (4.1)
E2 (linear trend)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.05	4.8 (6.2)	4.7 (17.0)	3.7 (27.0)	7.0 (4.4)
	0.1	5.2 (6.3)	4.9 (17.0)	3.7 (27.0)	7.6 (4.5)
	0.15	5.9 (6.4)	5.3 (17.0)	3.7 (27.0)	8.8 (4.6)
	0.2	7.0 (6.5)	6.0 (17.0)	3.7 (27.0)	10.4 (4.6)
E3 (adversarial block)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.05	4.8 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.1	4.9 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.15	5.2 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.2	5.5 (6.1)	4.7 (17.0)	4.2 (27.0)	6.8 (4.4)

Table 4: Full drift stress table at $\alpha = 0.1$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold**: realized flip exceeds α (certificate broken).

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	3.7 (5.1)	7.9 (5.0)	6.4 (7.0)	5.8 (4.4)
	0.05	5.6 (5.1)	9.1 (5.0)	7.8 (7.0)	7.1 (4.4)
	0.1	8.4 (5.2)	11.0 (5.0)	10.1 (7.0)	9.6 (4.3)
	0.15	11.9 (5.0)	12.7 (5.0)	12.3 (7.0)	12.2 (4.2)
	0.2	13.3 (4.7)	13.7 (5.0)	13.5 (7.0)	13.5 (4.1)
E2 (linear trend)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.5 (5.2)	8.9 (5.0)	7.5 (7.0)	7.0 (4.4)
	0.1	6.0 (5.3)	10.0 (5.0)	8.3 (7.0)	7.6 (4.5)
	0.15	6.9 (5.3)	11.6 (5.0)	9.7 (7.0)	8.8 (4.6)
	0.2	8.3 (5.4)	13.7 (5.0)	11.6 (7.0)	10.4 (4.6)
E3 (adversarial block)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.4 (5.1)	8.6 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.1	5.5 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.15	5.6 (5.1)	8.7 (5.0)	7.4 (7.0)	6.8 (4.4)
	0.2	5.7 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)

E THE CRITICAL-RADIUS CONSERVATION CURVE

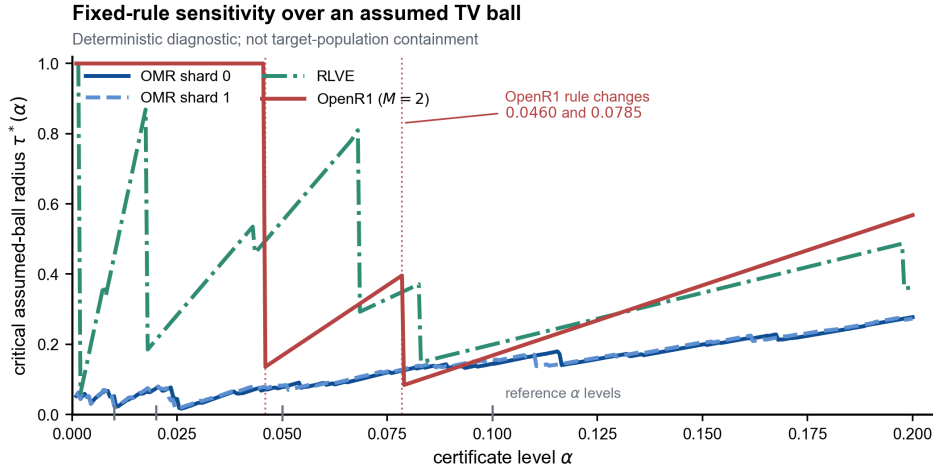


Figure 2: Deterministic fixed-rule sensitivity radius $\tau^*(\alpha)$ over an assumed fitted-mixture TV ball; this is not a confidence set for, or a containment claim about, an unknown target population. Proposition 1 is evaluated on the continuum $\alpha \in [10^{-3}, 0.2]$ (step 5×10^{-4}), computed exactly per level from the frozen g -profiles of the three carriers (OMR both FIT shards; OpenR1 exact in closed form). Grey tick marks: the four reference levels; red dotted lines: the OpenR1 certificate thresholds 0.0785/0.0460 where the frozen rule changes. Artifact: tv.conservation.r484.result.json.

Figure 2 makes the rule-change structure of the certificate visible. (a) *Staircase law*. Within any α -interval on which the frozen rule is unchanged, the whole vector g — hence $V(R)$ — is constant, so $\tau^*(\alpha) = \max\{R : V(R) \leq \alpha\}$ is *non-decreasing*; every strict decrease of τ^* coincides with a rule-change breakpoint (verified at every grid cell of every carrier; the rule-change counts on the plotted range are 96/93/7/2 for OMR shard 0 / OMR shard 1 / RLVE / OpenR1). (b) *Analytic breakpoints on OpenR1*. Its capacity-aware curve is $V(R) = g_{\max} \min\{1, \hat{H}(\frac{1}{2}) + R\}$; at the reported sub-capacity crossings, $\tau^*(\alpha) = \alpha/g_{\max} - \hat{H}(\frac{1}{2})$: at $\alpha = 0.07852$ the fail-prefix stop engages and $g_{\max}$ doubles ($0.125 \rightarrow 0.25$), so τ^* jumps down $0.396 \rightarrow 0.082$ ($\Delta = -0.314$); at $\alpha = 0.04598$ the pass-prefix stop disengages and the frozen rule degenerates to never-stop ($V(R) \equiv 0$ for any prior),

so the whole-simplex plateau $\tau^* = 1$ holds *below* this threshold and τ^* jumps *down* $1 \rightarrow 0.136$ ($\Delta = -0.864$) as α crosses it upward. (c) *Whole-simplex plateaus*. Wherever the frozen rule’s worst case cannot exceed α for *any* prior ($\max_K g(K) \leq \alpha$), $\tau^* = 1$: RLVE for $\alpha < 0.002$ and OpenR1 for $\alpha < 0.0460$; OMR’s rule never stops only below 10^{-3} , off the plotted range. (d) *Alignment co-variation (decomposed)*. The zero- g alignment number co-varies with τ^* at pooled Spearman 0.75 across the 16 carrier $\times$ level cells (descriptive; the per-cell alignment is itself α -dependent). The decomposition audit (spearman_decomposition_r486_result.json) shows this is a between-carrier law, not a within-carrier one: 1.0 across the four carrier means, 0.63 on the 14 non-plateau cells, 0.44–0.90 under leave-one-carrier-out; within-carrier it is $-0.95/-0.89$ (OMR shards), $+0.32$ (RLVE), $+1.0$ (OpenR1) — the OMR reversal is the staircase mechanism itself (level tightening adds zero- g mass on the lowest- g atoms only). Heterogeneity is not a confounder: het-excess vs τ^* Spearman 0.01, het-excess vs alignment 0.05 (16 cells). *Practical reading*. The staircase shape is a design signal for choosing the certificate level: inside a constant-rule interval, loosening α buys certified drift tolerance gradually; at a breakpoint it buys (or loses) a discrete jump — α should be set on a step, not inside one. (e) *How much FIT data buys how much robustness*. Rebuilding the prior from nested prefixes of the frozen FIT order at $m \in \{\frac{1}{32}, \dots, 1\} \times m_{\text{full}}$ and recomputing $\tau_m^*(\alpha)$ exactly (prior_fit_size_r491_result.json) gives an operational size rule. On OMR the $\tau_m^*(0.05)$ curves already clear the observed cross-shard transfer 0.042 at the smallest tested pool $m=125$ (shard 0: 0.062; shard 1: 0.046), i.e. cross-shard robustness does *not* require the full 4000-problem fit. The radius is *not* monotone in m (OMR shard 1 at $\alpha = 0.01$: 0.010 at $m=125$, up to 0.056 at $m=1000$, back to 0.024 at $m=4000$): growing the fit moves both the prior (better) and the frozen rule (larger g_{max} possible), and the rule channel can dominate — τ_m^* is a property of the fitted pair $(\hat{H}_m, \text{rule}_m)$, not of the sample size alone. The closed loop of Proposition 1 holds at every one of the 51 cells with $\text{TV}(\hat{H}_m, \hat{H}_{\text{full}}) \leq \tau_m^*$: the m -rule’s exact flip under the full-data prior never exceeds α (0 violations), while at the coarsest OMR shard 1 fit ($m=125$, TV 0.175, outside the ball) the transferred rule genuinely breaks (flip $0.0244 > 0.02$, $0.0165 > 0.01$) and is repaired from $m=250$ onward — inside/outside the radius predicts validity/failure exactly. The estimation error itself scales as $\text{TV}_m \approx C/\sqrt{m}$ (within-carrier spread of $\text{TV}_m \sqrt{m}$ below $1.5\times$ on OMR; the RLVE spread reaches $3.5\times$ only because its coarsest prefixes carry a missing-mode error $\sim \sqrt{m_{\text{full}}/m}$ larger). (f) *Which channel moves τ_m^* : rule, not prior — with one caveat*. We decompose the m -sensitivity into its two moving parts by computing, at every (carrier, m , α) cell, the two counterfactual radii (rule_channel_r494_result.json): $\tau_m^{*\text{pf}} = \tau^*(\hat{H}_{\text{full}}, g_m)$ freezes the prior at the full fit and lets only the *rule* vary with m , and $\tau_m^{*\text{rf}} = \tau^*(\hat{H}_m, g_{\text{full}})$ freezes the rule and lets only the *prior* vary. The recomputed both-moving curve reproduces the m -grid of prior_fit_size_r491_result.json bit-exactly (0/72 mismatches). Three findings. (i) *The rule channel drives the non-monotonicity*. With the prior frozen at $\hat{H}_{\text{full}}$, $\tau_m^{*\text{pf}}$ is still non-monotone at OMR shard 1 $\alpha=0.01$: 0.000 at $m=125$ rising to 0.060 at $m=1000$ then falling to 0.024 at $m=4000$ — the same up-then-down shape as the both-moving curve — because the refitted rule’s g_{max} itself varies non-monotonically ($0.563 \rightarrow 0.075 \rightarrow 0.098$) as the coarse prefix shifts which budgets are certifiable. (ii) *The estimation channel is benign on average but not uniformly*. Holding the rule fixed, a coarser prior can *shrink* the radius below the full-fit value in 31 of 72 cells (all on OMR shard 1 and RLVE; largest deficit 0.053 at RLVE $m=93$, $\alpha=0.05$). The mechanism is the base-flip term $\sum_K \hat{H}[K] g(K)$: a coarse prefix that under-covers the likely-correct (low- g) region and over-weights high- g mass raises the base flip, and — because the *frozen full rule* already has large g exactly on those atoms — the certificate has less slack to spend on drift. This is the missing-mode failure of (e) seen through the rule: the same coarse prefix that carries a $\sim \sqrt{m_{\text{full}}/m}$ estimation error also pays it on the worst (largest- g) coordinates. (iii) *The RLVE coarsest dip is pure rule-channel*. The collapse of the both-moving radius at RLVE $m=187$, $\alpha=0.05$ (0.235, far below both $m=93$ at 0.616 and $m=375$ at 0.534) survives with the prior frozen ($\tau_{187}^{*\text{pf}} = 0.222$) and vanishes with the rule frozen ($\tau_{187}^{*\text{rf}} = 0.530$): it is entirely a g_{max} event (0.179 vs. 0.071 for the full rule), not an estimation one. *Reading*. The robustness certificate is governed by the *rule* the fit induces more than by the prior’s sampling error: two fits of the same size can certify very different radii because the induced rule differs, and the only lever that reliably buys radius is one that shrinks the induced g -profile (e.g. the margin repair of Section 7), not one that merely adds FIT data. (g) *Oracle K -wise profile-capping sensitivity (withdrawn as a policy claim)*. The stored certificate-ordered subset calculation selects a separate set $S_K(c)$ after conditioning on each final count K and then evaluates the coordinate $g_{S_K}(K)$. It is therefore an *oracle K -wise profile-capping*

sensitivity calculation, not the flip profile of one rule. The final count K is unobserved at every nonterminal prefix, and the artifacts provide neither a common state table $S(x, k)$ nor a proof that their K-indexed memberships agree on shared reachable prefix states. Consequently, this calculation is not a stopping policy, not prefix-observable, and not an instance of Theorem 1. It cannot support executable or deployable repair, policy-level count/cost claims, or claims about extra votes, universal budgets, 70-of-72 cells, domination, edge laws, crossing sets, or full-simplex robustness. Those prior numerical interpretations are withdrawn; no new calculation is reported here. Establishing any policy-level repair would require one K-independent table $S(x, k)$ built only from observable prefix information, a prefix-measurability check, and recomputation of every profile and capacity-aware TV LP from that same table.

F CROSS-CARRIER EVIDENCE MATRIX AND UNIFIED MECHANISM

Table 5 collects every in-distribution TEST readout of the paper in one place: the four model-carrier pairs (dependence/resolvable structure decreasing from left to right), all stoppers, all certified levels, and one descriptive $\alpha=0.01$ CAL-side row for OMR (where the certified family of Section 5 reports only down to $\alpha=0.02$). Every flip in this matrix is replay agreement with a corresponding full vote and every saving is replay count reduction; it is not gold correctness or operational cost. Validity was certified on CAL (“cert”); TEST is a single descriptive readout under the Bonferroni radius (0.0286 OMR shard 0; 0.0292 shard 1; 0.0310 OpenR1; 0.0307 RLVE). Artifacts: `fit_cal_test_r469`, `shard1_robust_r471` (R1/R2), `openr1_m2_pilot_r473`, `rlve_n8_r474`.

Table 5: Replay evidence matrix: carrier (dependence level) $\times$ baseline $\times$ endpoint. “null” = no certifiable replay budget at that level; “—” = not applicable / not reported. FULL rows are the zero-count-reduction reference. Seeds 20260815 throughout.

Carrier	rule	α	cert	replay flip	mean k	count reduction
OMR s0 ($N=32$)	FIXED-HOEF	.10	0.0970	0.0645	7.0	78.1%
	FIXED-EB	.10	0.0958	0.0794	5.0	84.4%
	BAYES-H	.10	0.0495	0.0370	5.1	84.0%
	FIXED-HOEF	.05	0.0471	0.0156	27.0	15.6%
	FIXED-EB	.05	0.0461	0.0324	17.0	46.9%
	BAYES-H	.05	0.0350	0.0249	6.1	80.9%
	FIXED-HOEF	.02	null	—	—	—
	FIXED-EB	.02	0.0171	0.0083	31.0	3.1%
	BAYES-H	.02	0.0163	0.0091	8.5	73.5%
	BAYES-H	.01	0.0119	—	—	—
	FULL-32	—	—	0	32.0	0
OMR s1 ($N=32$)	FIXED-HOEF	.05	0.0488	0.0171	25.0	21.9%
	FIXED-EB	.05	0.0486	0.0357	15.0	53.1%
	BAYES-H	.05	0.0340	0.0246	6.2	80.6%
	BAYES-H [†]	.05	0.0341	0.0247	6.2	80.6%
	BAYES-H	.02	0.0158	0.0088	8.6	73.1%
	BAYES-H [†]	.02	0.0159	0.0089	8.6	73.2%
OpenR1 ($M=2$)	FIXED-1	—	0.0707	0.0592	1.0	50.0%
	BAYES-H	.10	0.0707	0.0592	1.0	50.0%
	BAYES-H	.05	0.0383	0.0294	1.4	32.1%
	BAYES-H	.02	0.0047	0	2.0	0
RLVE ($N=8$)	FIXED-HOEF	.10	0.0942	0.0613	5.0	37.5%
	FIXED-EB	.10	0.0783	0.0613	5.0	37.5%
	BAYES-H	.10	0.0469	0.0314	3.4	57.6%
	FIXED-EB/HOEF	.05-.01	null	—	—	—
	BAYES-H	.05	0.0184	0.0103	3.9	50.9%
	BAYES-H	.02	0.0121	0.0046	4.1	48.2%
	BAYES-H	.01	0.0081	0.0019	4.4	45.4%

[†]Cross-shard prior transfer (prior fit on shard 0 FIT, TV=0.042 to the shard 1 prior); all numbers move ≤ 0.4 points against within-shard. WINDOW3 (no certificate knob) is omitted from the matrix: it has the largest replay count reduction (86.2%) but is invalid at $\alpha \leq 0.05$ (flip 0.0577; Table 1).

Unified mechanism. One quantity — the prior’s g -profile — organizes every panel of the paper. (i) *In-distribution*, replay count reduction tracks the resolvable mixture structure: mass on extreme- K problems stops at $k \approx 5$ (OMR, 81% reduction at $\alpha=0.05$); with only $N=8$ draws and 74–87% of prior mass on zero- g atoms, three stoppable budgets remain and the count reduction interpolates to 45–58% (RLVE); at $M=2$ the single stoppable budget yields 32–50% and the certificate degenerates to never-stop exactly when $\max_K g(K) \leq \alpha$ (OpenR1 at $\alpha=0.02$: count reduction 0, flip 0 — the certificate never over-stops). (ii) *Under prior shift*, the same profile sets the critical radius: τ^* is increasing in the zero- g atom mass across carriers (Spearman 1.0 on carrier means, 16 cells; het-excess variance is not the driver: RLVE 0.0999 < OMR 0.129 yet tolerates more), with the within-OMR negative trend (−0.95/−0.89) a staircase artifact (decomposition audit, `spearman_decomposition_r486_result.json`). (iii) *Under synthetic ordered drift*, adaptivity is itself the robustness channel in the specified harness: BAYES-H re-reads the accumulating evidence each draw, so it is the most robust stopper under the smooth E2/E3 mechanisms (valid to $\delta=0.20$ under E2 at $\alpha=0.10$ where FIXED- k breaks at 0.15), and a margin γ on the certificate buys flip control at a quantified replay-count cost ($\gamma=0.03$ covers E2 $\delta \leq 0.20$: flip 0.0491, count reduction 71.8% vs 79.7% unmargined; $\gamma=0.025$ covers E3 $\delta \leq 0.15$: flip 0.0495, count reduction 77.2% vs 81.0%; `margin_repair_r469_result.json`). (iv) *The failure boundary is sharp on every axis*: premise-breaking E1 breaks all certificates together by $\delta=0.10$ –0.15; on shard 1 at $\alpha=0.01$ the observed cross-shard mismatch 0.042 exceeds $\tau^*(0.01)=0.024$ and TV alone does not certify the transfer; the coarsest fit $m=125$ sits outside its ball (TV 0.175) and genuinely breaks at $\alpha \leq 0.02$ (Appendix E(e)). Where the paper claims a win, the margin is large and certified; where it does not, the limiting quantity is named and measured.

G REPRODUCIBILITY

All numbers trace to JSON artifacts listed in REPRO_README.md; carrier SHA-256 pinned; deterministic seeds; CPU-only; total compute under 25 minutes.